FullonBaan
— One stop destination for IT solutions —

ONE STOP DESTINATION FOR IT SOLUTIONS

OFFICIAL TRAINING GUIDE

Architecture

Infor ERPLN 6.1 Tools

Architecture | Session Management | System Tiers | VRC Concepts

PREMIUM EDITION

© 2025 FullonBaan IT Solutions

LEGAL NOTICE & COPYRIGHT STATEMENT

COPYRIGHT © 2025 FULLONBAAN IT SOLUTIONS

ALL RIGHTS RESERVED

RESTRICTED DOCUMENT – CONFIDENTIAL TRAINING MATERIAL

This document, including all text, diagrams, illustrations, and associated content contained herein, is the exclusive intellectual property of FullonBaan IT Solutions and is protected under applicable copyright law, international treaties, and intellectual property regulations.

REDISTRIBUTION STRICTLY PROHIBITED: No part of this document may be reproduced, photocopied, stored in a retrieval system, transmitted in any form or by any electronic or mechanical means, posted to any public or private network or platform, shared, distributed, sublicensed, resold, or otherwise transferred to any third party without the express prior written consent of FullonBaan IT Solutions.

AUTHORIZED USE ONLY: This material is licensed solely for use by the authorized recipient for personal professional development and internal training purposes. Any unauthorized use constitutes a violation of copyright law and may result in civil and/or criminal penalties.

For licensing inquiries, contact: legal@fullonbaan.com

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

Table of Contents

01 **Software Architecture**
Sessions, Forms, Tables, Reports, Charts, Objects

02 **Session Startup Flow**
Step-by-step lifecycle from launch to user control

03 **System Architecture & Tiers**
UI, Application Logic, Database tiers & deployments

04 **Data Dictionary**
Runtime DD, Application DD, Convert to Runtime

05 **Version Release Customization (VRC)**
VRC concept, terminology, Package VRC & combinations

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

Lesson Objectives

Upon completing this module, you will be able to:

01 Identify the components of a session

02 Describe major steps when a user starts a session

03 Identify the tiers of the Infor ERP system architecture

04 Describe major steps when a user starts the Infor ERP software

05 Explain the VRC concept

06 Find out from which VRC a session component will be retrieved

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

SECTION 01

Software Architecture

Understanding Sessions, Forms, Tables, Reports, Charts & Objects

Architecture Overview

Tools will determine how the end user uses the system.

Software Architecture

DEFINES

How a session starts — defines the software components that make up a session, including forms, tables, objects, reports and charts.

System Architecture

DEFINES

How the system starts — defines the technical environment and tiers on which the Infor ERP application runs.

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

Structure of a Session

A Session performs a specific activity within the Infor ERP system. Each session is composed of five key components:

OBJ

Object

Contains most of the business logic and controls tasks performed by the session. Can access DLLs for common functions.

FRM

Forms

The User Interface part of the session. Presents data to users, identifies fields, labels and options in Overview and Details windows.

TBL

Tables

Defines the data structure. Tables have Fields, Fields have Data Types, and Fields can reference related tables.

RPT

Reports

Print or display information from the database. Initiated via a print session or the printer icon.

CHT

Charts

Present data in a graphical format, enabling visual analysis of business data stored in the system.

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

Forms & Table Definitions

FORMS

- Form is the User Interface part of the session
- Forms are used to interact with users by presenting data
- Session and Form are tightly integrated
- Form Definition identifies fields, labels and options in both Overview and Details session windows
- Overview Sessions show multiple rows in a grid; Details Sessions show one row at a time

TABLE DEFINITIONS

- Defines the structure of data within the ERP system
- Tables contain Fields which define data attributes
- Each Field has an associated Data Type
- Fields can reference and link to related tables
- Table data is accessed and displayed through Sessions

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

Reports, Charts & Objects

REPORTS

- Print or display information from the database
- Initiated via a print session selection
- Also accessible through the printer icon in the toolbar
- Used for output and analysis of ERP data

CHARTS

- Present data in a graphical and visual format
- Enable trend analysis and business insights
- Linked to session data and updated dynamically
- Support bar, line, pie and other chart types

OBJECT

- Contains the core business logic of the session
- Controls all tasks performed within a session
- Can access Dynamic Link Libraries (DLLs)
- DLLs provide reusable common functionality

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

SECTION 02

Session Startup Flow

Step-by-step lifecycle from launch to passing control to the user

Session Startup Flow

1. **Query runtime components**
   Form, object, main table, reports...
2. **Load session object**
   Business logic object is loaded
3. **Check license**
   Valid license verified
4. **Create session window**
   A new thread is created
5. **Load standard components**
   ottstp_stddll, ottstpstandard...
6. **Check session authorizations**
   User permission check
7. **Load form & present tab**
   First form tab displayed in window
8. **Load data definitions**
   Main table and related tables
9. **Load data management**
   Integration components loaded
10. **Query & select data**
    Data retrieved from tables
11. **Apply user defaults**
    Preset form data from user profile
12. **Pass control to user**
    Session fully handed to end user

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

SECTION 03

System Architecture & Tiers

Three-tier design: Presentation Logic, Business Logic & Data Management

Three-Tier System Architecture

Infor ERP separates responsibilities into three distinct architectural tiers:

TIER 1

PRESENTATION LOGIC

USER INTERFACE

- Handles all user interface interactions
- Window placement and form painting
- Mouse control and user input
- Worktop: runs directly on a PC
- Webtop: runs on web server, accessed via browser

TIER 2

BUSINESS LOGIC

APPLICATION

- The bshell virtual machine processor
- Similar in concept to a Java Virtual Machine
- Middleware: Authentication, Security, Data Connection
- Runs on the application server
- Processes all session business rules

TIER 3

DATA MANAGEMENT

DATABASE

- Manages all business information
- Object sends generic SQL via database driver
- Driver translates to DB-specific SQL commands
- Compatible with multiple database engines
- Data persisted in relational database tables

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

Deployment Scenarios

Infor ERP supports multiple deployment configurations to suit various infrastructure needs:

SINGLE HOST

All tiers (Presentation, Application, Database) run on one server. Clients connect via Company LAN. Suitable for smaller installations.

- Laptop, PCs and Printers on Company LAN
- Single Server hosts Application (A) + Database (D)
- Presentation (P) runs on each client workstation

DISTRIBUTED

Application and Database tiers separated onto dedicated servers. A Server-Only LAN connects backend servers for improved performance and security.

- Separate Application Server and Database Server
- Server-Only LAN connects backend servers
- Clients connect via Company LAN to Application Server

SINGLE HOST WITH WEBTOP

Webtop enables browser-based access. A Web Server handles Presentation logic; clients need only a web browser — no Worktop client install required.

- Web Server handles Presentation (P) logic
- Backend Server hosts Application (A) + Database (D)
- Clients access via browser — no client install needed

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

System Logon & Startup Flow

1. **Start user interface program**
   Worktop or Webtop launched
2. **Logon to system**
   User credentials authenticated
3. **Start application services (bshell)**
   Virtual machine processor initialized
4. **Load environment defaults**
   System-wide settings loaded
5. **Load user profile & settings**
   Personal preferences applied
6. **Load user interface settings**
   UI configuration loaded
7. **Start database driver**
   DB connection initialized
8. **Start standard objects & libraries**
   Core libraries loaded
9. **Check license**
   License validity confirmed
10. **Query user interface data**
    Navigation data retrieved
11. **Start navigation session**
    User's main menu presented

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

SECTION 04

Data Dictionary

Runtime Data Dictionary, Application Data Dictionary & Convert to Runtime

Data Dictionary Overview

A data dictionary is a collection of descriptions about a data model or system. Infor ERP uses two unique data dictionaries:

APPLICATION DATA DICTIONARY

APD

D

Data stored in tables, accessed and managed by the Tools sessions within Infor ERP. This is the source of truth for application metadata.

- Data stored in relational database tables
- Accessed and edited via Tools sessions
- Defines sessions, forms, tables, objects
- Source for Convert to Runtime process

Convert
to Runtim

RUNTIME DATA DICTIONARY

RTDD

Files and data used to run the system at runtime. Serves a dual purpose — provides technical configuration and optimized application information.

- Provides technical architecture configuration
- Delivers optimized application runtime data
- Created via Convert to Runtime process
- Consumed by bshell during session startup

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

Convert to Runtime Data Dictionary

The process to make APDD available in the RTDD is called "Convert to Runtime". This step must be performed after any changes to application definitions.

Application Data Dictionary (APDD)

Data in tables, managed by Tools sessions

→

Convert / Compile or Dump

Triggered via Convert to Runtime sessions

→

Runtime Data Dictionary (RTDD)

Optimized files used at runtime by bshell

Convert to Runtime Sessions

ttams2200m000	Convert Changes to Runtime DD
ttams2201m000	Convert to Runtime DD
ttadv5215m000	Convert to Runtime Data Dictionary
ttadv5210m000	Create Runtime Data Dictionary

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

SECTION 05

Version Release Customization

VRC concept, Package VRC, Package Combination & software lifecycle management

The VRC Concept

VRC (Version Release Customization) manages component changes within the Infor ERP system:

VERSION

A major change to the software — represents a significant architectural or functional overhaul.

RELEASE

A minor change to the software — incremental updates, patches, and small enhancements.

CUSTOMIZATION

A customer-specific change — tailored modifications unique to a particular implementation.

Why VRC?

Software Version Control
Control exactly which version of the software runs in production at any time.

Developer Isolation
Segregate development environments from live production environments safely.

Upgrade Testing
Test a software upgrade in parallel and promote it to production when ready.

Multi-user Flexibility
Different users can run different versions of the same session component simultaneously.

© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

VRC Terminology & Package VRC

VRC TERMINOLOGY

Package

A set of related software components that belong together as a logical unit (e.g., tc for common, wh for warehouse).

Package VRC

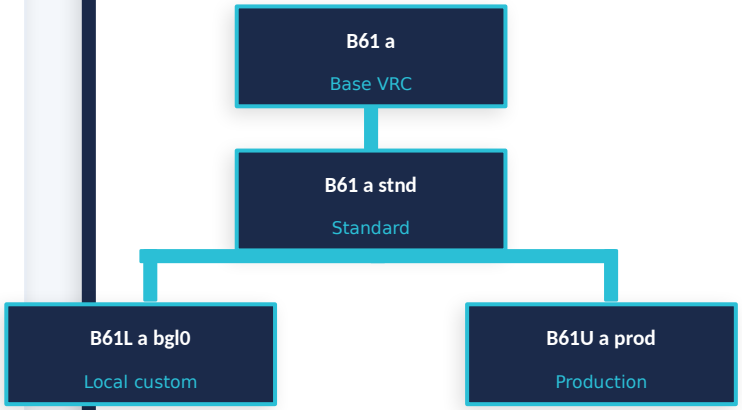
A unique version of a package — identified by a version, release, and customization identifier string.

Package Combination

A group of packages assembled together with specified versions, forming a complete ERP configuration.

PACKAGE VRC DERIVATION

A VRC can be derived from another VRC. A derived VRC only contains new or changed components that differ from its parent — unchanged components are inherited from the parent VRC.



© 2025 FullonBaan IT Solutions. All rights reserved. Confidential.

One stop destination for IT solutions

This concludes the Architecture module for Infor ERPLN 6.1 Tools.

www.fullonbaan.com

Module Complete

Architecture | Infor ERPLN 6.1 Tools

Topics covered in this module:

- Software Architecture & Session Components
- Session Startup Flow
- Three-Tier System Architecture
- Deployment Configurations
- Data Dictionary (APDD & RTDD)
- Version Release Customization (VRC)